

Solving Optimal Power Flow Problem in Hybrid Renewable Energy Systems Through Hybrid Optimization Algorithm

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Abstract

The growing global emphasis on sustainability has accelerated the transition from fossil fuels to renewable energy sources. Hybrid renewable energy systems, integrating wind and solar energy, have emerged as efficient alternatives for cleaner energy generation. However, managing Optimal Power Flow (OPF) within these systems remains a critical challenge due to the inherent uncertainties in renewable energy resources and the nonlinear nature of OPF problems. This study proposes a novel hybrid optimization approach—hybrid spotted hyena optimization algorithm—that integrates spotted hyena optimization with a quadratic approximation operator and grasshopper optimization to

improve convergence speed, solution quality, and resilience to uncertainties.

Uncertainties in wind and solar generation are modeled using Weibull and log-normal probability distributions, respectively, incorporated through Monte Carlo simulations. The algorithm is tested on the modified IEEE-30 bus system. Compared to conventional methods like PSO, GA, and GWO, the proposed model achieved a minimum fuel cost of 658.98 USD/hr, reduced power loss to 3.684 MW, and minimized voltage deviation to 0.0756 p.u., demonstrating significant improvements in efficiency and reliability. These results confirm the broader applicability of the model in achieving sustainable and cost-effective power management in renewable-driven smart grids.

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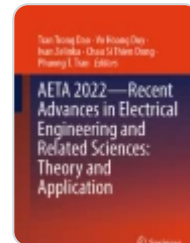
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Availability of Data and Materials

Data is provided within the manuscript.

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Contributions

All the authors contributed to this research work in terms of concept creation, conduct of the research work, and manuscript preparation.

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Ethics declarations

Conflict of interest

The authors declare no competing interests.

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Appendix

See Tables ([9](#), [10](#), [11](#) and [12](#))

Table 9 Active power generation limits for generators

Table 10 Voltage magnitude limits at generator buses

Table 11 Tap settings of adjustable transformers

Table 12 Shunt VAR compensation limits

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